

DATA STRUCTURE

LAB PROGRAMS

17. Insertion sorting .

```
#include <stdio.h>

int main()
{
    int arr[] = {9, 5, 1, 4, 3};
    int n = 5;
    for(int i = 1; i < n; i++)
    {
        int key = arr[i];
        int j = i - 1;
        while(j >= 0 && arr[j] > key)
        {
            arr[j + 1] = arr[j];
            j--;
        }
        arr[j + 1] = key;
    }
    printf("Sorted Array: ");
    for(int i = 0; i < n; i++)
        printf("%d ", arr[i]);
    return 0;
}
```

Output

Sorted Array: 1 3 4 5 9

=== Code Execution Successful ===